MNIST AI Kurs uppgifter

Innehållsförteckning

mnist-start
mnist-mlp
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mnist-start

```
    Directory: G:\Min enhet\AI_prov\mnist-start


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        2026-05-09     10:26                .ipynb_checkpoints
d-----        2026-05-09     10:27                data
------        2026-05-09     10:01         490250 Mnist - funderingar.pptx
------        2026-05-08     08:39         411816 Göran_ML_Flöde_kod.ipynb
------        2026-05-09     10:42         275355 Göran Mnist version 10.ipynb
------        2026-05-07     19:00           5187 Göran_ML_flöde.ipynb
------        2026-05-09     10:58         316158 MNIST Read Me.pdf
------        2026-04-03     11:06           5509 digit5.png
```

mnist-mlp

```
    Directory: G:\Min enhet\AI_prov\mnist-mlp


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        2026-05-09     07:17                Bak
------        2026-05-09     07:35           2205 Göran_ML_MLP_sparad.py
------        2026-05-09     07:57           1756 Göran_ML_MLP_modell.py
------        2026-05-09     07:58              6 accuracy.txt
------        2026-05-09     07:58        1637792 mnist_mlp.pkl
```

Steg 1
python Göran_ML_MLP_modell.py

```
(py311) PS G:\Min enhet\AI_prov\mnist-mlp> python Göran_ML_MLP_modell.py
Laddar MNIST...
Model: MLPClassifier
Tränar MLP...
Accuracy: 0.9738
Modell sparad som mnist_mlp.pkl
Modell och accuracy sparad.
```

Steg 2
streamlit run Göran_ML_MLP_sparad.py

```
(py311) PS G:\Min enhet\AI_prov\mnist-mlp> streamlit run Göran_ML_MLP_sparad.py
2026-05-09 07:58:32.168 Uvicorn server started on 0.0.0.0:8501

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://192.168.0.184:8501

  Stopping...
(py311) PS G:\Min enhet\AI_prov\mnist-mlp>
```

MNIST MLP Digit Recognizer

Model: MLPClassifier

Model Accuracy: 97.38%

Rita en siffra mellan 0 och 9

Gissning: 3

Sannolikheter

0: 0.00%

1: 0.00%

2: 0.00%

3: 100.00%

mnist-cnn

```
    Directory: G:\Min enhet\AI_prov\mnist-cnn


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        2026-05-09     07:17                bak
------        2026-05-08     16:33              6 accuracy.txt
------        2026-05-08     16:33        2740015 mnist_cnn.keras
------        2026-05-08     10:11           2355 Göran_ML_CNN_model.py
------        2026-05-08     16:37           3176 Göran_ML_CNN_sparad.py
```

Steg 1
python Göran_ML_CNN_model.py

```
(py311) PS G:\Min enhet\AI_prov\mnist-cnn> python Göran_ML_CNN_model.py
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN
are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
Epoch 1/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.9379 - loss: 0.2027 - val_accuracy: 0.9868 - val_loss: 0.0487
Epoch 2/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9796 - loss: 0.0656 - val_accuracy: 0.9883 - val_loss: 0.0361
Epoch 3/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9857 - loss: 0.0473 - val_accuracy: 0.9882 - val_loss: 0.0408
Epoch 4/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9887 - loss: 0.0382 - val_accuracy: 0.9918 - val_loss: 0.0305
Epoch 5/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9906 - loss: 0.0293 - val_accuracy: 0.9918 - val_loss: 0.0286
Epoch 6/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9919 - loss: 0.0245 - val_accuracy: 0.9925 - val_loss: 0.0292
Epoch 7/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9932 - loss: 0.0211 - val_accuracy: 0.9923 - val_loss: 0.0297
Epoch 8/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9940 - loss: 0.0175 - val_accuracy: 0.9922 - val_loss: 0.0319
Epoch 9/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9944 - loss: 0.0157 - val_accuracy: 0.9925 - val_loss: 0.0322
Epoch 10/10
844/844 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.9953 - loss: 0.0135 - val_accuracy: 0.9920 - val_loss: 0.0317
313/313 ━━━━━━━━━━━━━━━━━━━━ 0s 2ms/step - accuracy: 0.9918 - loss: 0.0244
Accuracy: 0.9918
Modell och accuracy sparad.
CNN-modell sparad.
```

Steg 2
streamlit run Göran_ML_CNN_sparad.py

MNIST CNN Digit Recognizer

Model Accuracy: 99.18%

Rita en siffra mellan 0 och 9

Gissning: 3

Sannolikheter

0: 0.00%

1: 0.53%

2: 2.28%

3: 97.11%

4: 0.08%

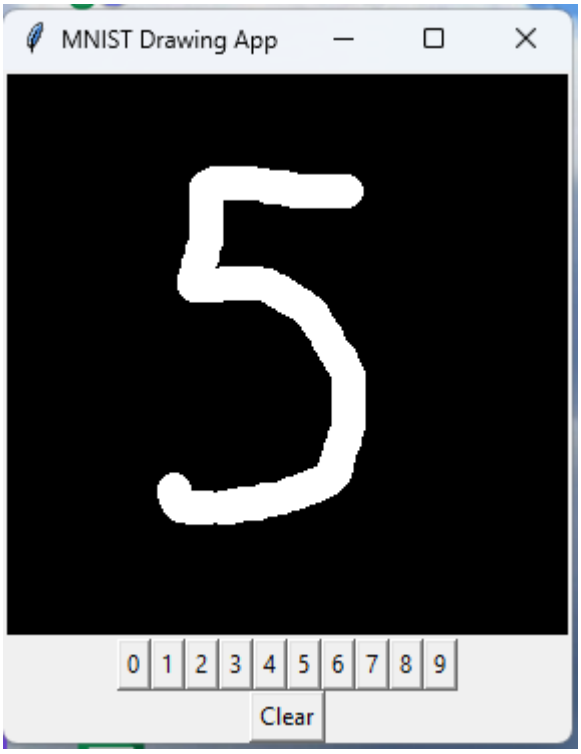
mnist-pics

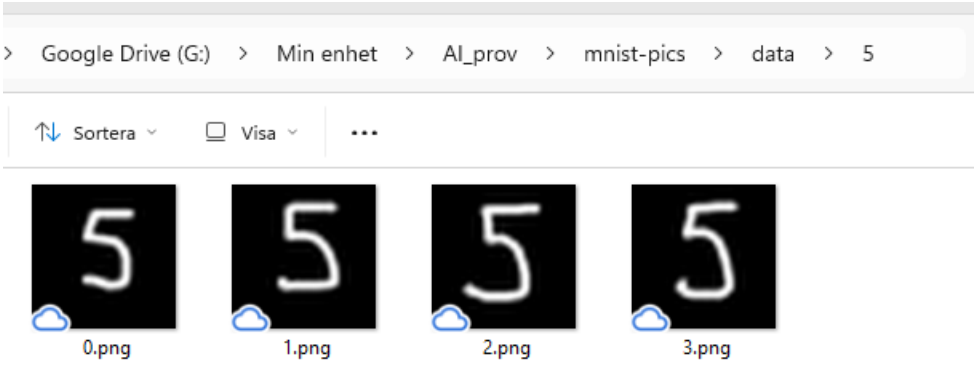
```
    Directory: G:\Min enhet\AI_prov\mnist-pics


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        2026-05-09     09:14                data
------        2026-05-09     09:14           1980 mnist-pics.py


(py311) PS G:\Min enhet\AI_prov\mnist-pics>
```

python mnist-pics.py





:orn > Google Drive (G:) > Min enhet > AI_prov > mnist-pics > data >

Sortera Visa ...

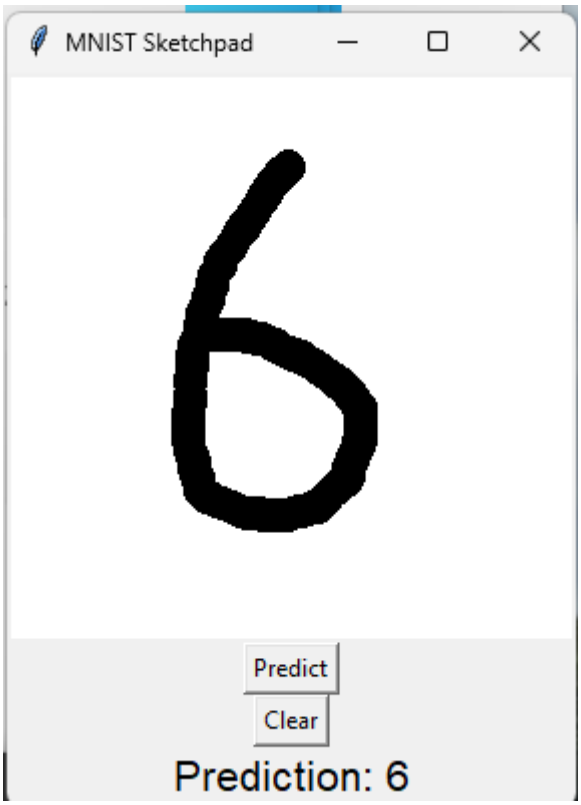
Namn	Senast ändrad	Typ	Storlek
0	2026-05-09 09:14	Filmapp	
1	2026-05-09 09:16	Filmapp	
2	2026-05-09 09:14	Filmapp	
3	2026-05-09 09:14	Filmapp	
4	2026-05-09 09:24	Filmapp	
5	2026-05-09 09:25	Filmapp	
6	2026-05-09 09:14	Filmapp	
7	2026-05-09 09:14	Filmapp	
8	2026-05-09 09:14	Filmapp	
9	2026-05-09 09:14	Filmapp	

mnist-prediktion

mnist-prediktion.py

Test av prediktionsförmåga.
Modell CNN

```
    Directory: G:\Min enhet\AI_prov\mnist-prediktion


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
------        2026-05-09     10:37        1500619 mnist_cnn.keras
------        2026-05-09     10:50           1684 mnist-prediktion.py
```

python mnist-prediktion.py